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DATA HANDLING AND VISUALIZATION

CO1 – AT3

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COURSE	DATA HANDLING AND VISUALIZATION
ASSESSMENT	CO1 – AT3

Overview and Assessment Strategy

This report presents a complete R-only visualization design solution for all four questions in the DAS0601 Data Handling and Visualization C01–AT3 exercise. The emphasis is not only on producing graphs, but also on demonstrating the complete visualization design process: understanding the structure of the data, selecting suitable visual encodings, applying transformations or coordinate systems where required, creating readable plots in RStudio, and explaining the analytical insight obtained from the final visualizations.

General objectives

- Import and structure tabular data in R using data frames/tibbles.
- Select visual encodings that match the type and scale of each variable.
- Use logarithmic transformation when a variable is strongly right-skewed.
- Build an integrated multi-view visualization for a multidimensional business dataset.
- Compare Cartesian and polar coordinate systems using the same seasonal energy dataset.
- Use multiple aesthetics such as position, colour, size, shape and faceting without creating unnecessary visual complexity.
- Interpret patterns, trends, relationships, outliers and negative values from the resulting plots.
- Document the visualization decisions clearly enough for reproduction in RStudio.

Design principles applied

Principle	Application in this report
Data-to-visual mapping	Quantitative variables are primarily mapped to position; categorical variables are mapped to colour/fill/facet.
Scale selection	Logarithmic scaling is used for the strongly skewed LinesOfCode variable in Q1.
Comparability	Common axes, labels, legends and units

	are used so that categories can be compared correctly.
Readability	Minimal themes, informative titles, captions and sensible number formatting are used.
Interaction of aesthetics	Size, colour and shape are used only when they add a distinct analytical meaning.
Reproducibility	All plots are produced from explicit R code so the student can rerun them in RStudio.

Common RStudio Setup

Run this setup once before running the four question-specific sections.

```
# Install packages once if they are not already installed:
install.packages(c("ggplot2", "dplyr", "scales", "patchwork"))

# Load packages for the current RStudio session:
library(ggplot2)
library(dplyr)
library(scales)
library(patchwork)

# Optional: clear the workspace
rm(list = ls())
```

If a package is already installed, R may report that no action is necessary. The installation command is normally required only once; the `library()` commands are used whenever a new R session starts.

Recommended RStudio workflow

1. Open RStudio and create a new R script (File → New File → R Script).
2. Copy the setup code and run it.
3. For each question, create the data frame exactly as shown.
4. Run the plotting code line by line or select the complete block and click Run.
5. Check the Plots pane for the generated visualization.
6. Use Export → Save as Image/PDF if a separate file is required.
7. Paste the final plot screenshot in the placeholder provided in this report.

Q1 – Nonlinear Axis for Skewed Data

Problem focus: A software company has recorded project statistics for twenty software development teams. The task requires a visualization using a nonlinear axis transformation so that the selected transformation and explanation highlight patterns that may otherwise be hidden.

Dataset and variables

Variable	Type	Meaning
Team	Categorical	Team identifier T1–T20
LinesOfCode	Quantitative	Number of lines of code
Bugs	Quantitative	Recorded number of bugs
Devs	Quantitative	Number of developers
Complexity	Quantitative	Complexity score

The LinesOfCode values increase from 12,000 to 1,800,000. This large range makes a linear x-axis unsuitable for showing the smaller teams with enough visual separation.

Visualization design

Visual property	Mapping	Reason
X-position	LinesOfCode	Primary explanatory variable; displayed on a logarithmic scale.
Y-position	Complexity	Shows how complexity changes as project size increases.
Point size	Bugs	Shows the relative bug count without requiring a second chart.
Colour	Devs	Shows the relative number of developers.

Label	Team	Identifies individual teams in the final plot.
Scale	log10 x-axis	Prevents small values from being compressed near zero.

3.3 R code – data entry

```
q1 <- data.frame(
  Team = paste0("T", 1:20),
  LinesOfCode = c(
    12000, 18000, 25000, 35000, 48000,
    62000, 81000, 96000, 120000, 150000,
    185000, 240000, 310000, 390000, 480000,
    620000, 810000, 1050000, 1350000, 1800000
  ),
  Bugs = c(
    35, 42, 60, 72, 88,
    110, 125, 142, 165, 188,
    205, 240, 280, 320, 365,
    420, 490, 560, 640, 720
  ),
  Devs = c(
    5, 7, 9, 10, 12,
    14, 16, 18, 20, 22,
```

```
25, 28, 30, 34, 38,  
42, 46, 50, 55, 60  
,  
Complexity = c(  
2.1, 2.4, 2.7, 3.0, 3.2,  
3.5, 3.8, 4.1, 4.5, 4.8,  
5.1, 5.4, 5.8, 6.2, 6.6,  
7.1, 7.6, 8.0, 8.5, 9.0  
)  
)
```

R code – nonlinear visualization

```
q1_plot <- ggplot(  
q1,  
aes(  
x = LinesOfCode,  
y = Complexity,  
size = Bugs,  
colour = Devs,  
label = Team  
)  
) +
```

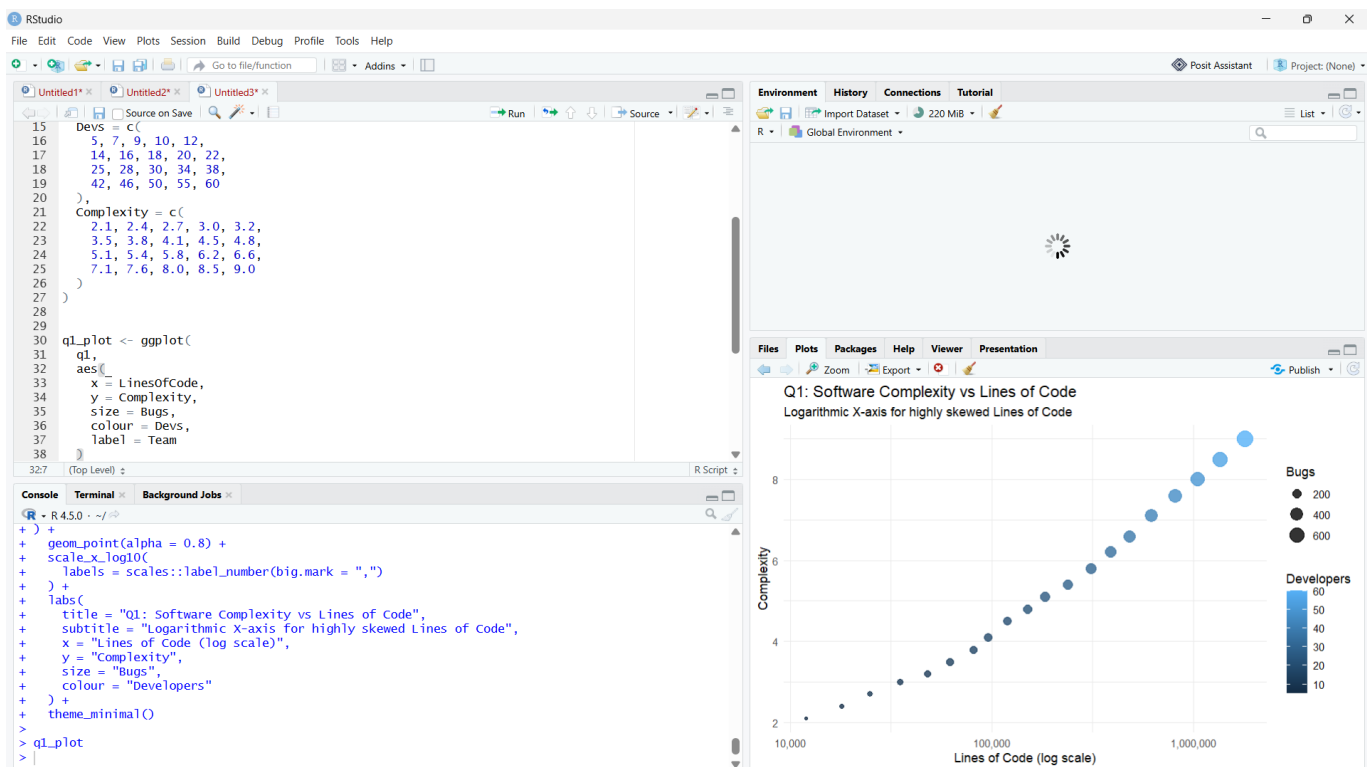
```
geom_point(alpha = 0.8) +  
scale_x_log10(  
  labels = scales::label_number(big.mark = ",")  
) +  
labs(  
  title = "Q1: Software Complexity vs Lines of Code",  
  subtitle = "Logarithmic X-axis for highly skewed Lines of Code",  
  x = "Lines of Code (log scale)",  
  y = "Complexity",  
  size = "Bugs",  
  colour = "Developers"  
) +  
theme_minimal()  
  
q1_plot
```

Why the logarithmic transformation is appropriate

A logarithmic axis changes the visual interpretation from equal absolute increments to equal multiplicative increments. For example, the distance from 10,000 to 100,000 is treated similarly to the distance from 100,000 to 1,000,000 because both represent a ten-fold increase. This is appropriate when the variable spans several orders of magnitude. In this dataset, a linear scale would place most observations near the left side because the largest projects are much larger than the smallest projects.

Interpretation points to report

- The plot allows both small and large software teams to remain visually distinguishable.
- Complexity generally increases as LinesOfCode increases across the ordered teams.
- Bugs also rise substantially toward the larger projects, so point size becomes larger at the right side.
- Developer count increases with project size, which is visible through the colour scale.
- The log transformation does not change the underlying data values; it changes the axis scale so relative growth can be visualized more effectively.
- Team labels provide traceability, but if the final screenshot becomes crowded, labels may be removed and team information retained in tooltips or a supporting table.



conclusion

The logarithmic x-axis is the key design decision. It directly addresses the skewness in LinesOfCode and produces a more informative view of the relationship between project size and complexity while allowing Bugs and Developers to be incorporated as additional aesthetics.

Q2 – Integrated / Capstone Visualization Design

Problem focus: A financial analytics company has performance data for twenty companies across IT, Banking, Healthcare, Retail and Energy. The visualization must communicate Revenue, Profit Margin, Market Capitalization, Employees and Sector in an integrated design.

Variables

Variable	Type	Role in visualization
Company	Categorical	Company identifier C1–C20
Sector	Categorical	Industry group
Revenue	Quantitative	Business scale
ProfitMargin	Quantitative	Profitability percentage
Employees	Quantitative	Organizational size
MarketCap	Quantitative	Market valuation

Integrated dashboard concept

Instead of forcing all five variables into one crowded plot, the solution uses several coordinated panels. The panels share the same Sector colour mapping so the viewer can move between views while retaining category identity.

Panel	Chart	Main variables	Purpose
A	Scatter plot	Revenue × ProfitMargin; size = MarketCap; colour = Sector	Shows business scale, profitability and valuation together.
B	Bar chart	Revenue by Company; fill = Sector	Provides a direct comparison of company revenue.
C	Scatter plot	Employees × MarketCap; colour	Shows whether workforce size and

		= Sector; size = ProfitMargin	valuation move together.
D	Boxplot	ProfitMargin by Sector	Summarizes profitability distribution across sectors.

4.3 R code – data entry and derived variables

```
q2 <- data.frame(
  Company = paste0("C", 1:20),
  Sector = c(
    "IT", "Banking", "Healthcare", "Retail", "Energy",
    "IT", "Banking", "Healthcare", "Retail", "Energy",
    "IT", "Banking", "Healthcare", "Retail", "Energy",
    "IT", "Banking", "Healthcare", "Retail", "Energy"
  ),
  Revenue = c(
    420, 580, 720, 850, 980,
    1150, 1320, 1480, 1650, 1820,
    2050, 2280, 2550, 2840, 3150,
    3480, 3860, 4250, 4680, 5200
  ),
  ProfitMargin = c(
    8.2, 15.6, 11.4, 9.3, 18.5,
```

```

20.4, 17.8, 14.2, 10.5, 22.1,
24.5, 19.8, 16.7, 13.4, 26.2,
28.1, 25.4, 18.3, 15.2, 30.4
),
Employees = c(
1500, 3200, 2100, 4200, 2800,
5100, 6800, 3900, 7200, 4500,
8100, 9500, 6200, 10300, 7200,
11800, 13200, 8600, 14500, 16000
),
MarketCap = c(
8500, 12000, 9800, 7600, 14500,
18200, 21500, 16800, 13200, 24500,
28500, 32500, 24800, 19200, 38200,
42500, 46800, 35600, 29800, 52000
)
)
)

```

4.4 R code – integrated dashboard

```

q2_plot <- ggplot(
  q2,
  aes(

```

```

  x = Revenue,
  y = MarketCap,
  size = Employees,
  colour = Sector
)
) +
geom_point(alpha = 0.8) +
labs(
  title = "Q2: Integrated Corporate Performance",
  x = "Revenue",
  y = "Market Capitalization",
  size = "Employees",
  colour = "Sector"
) +
theme_minimal()

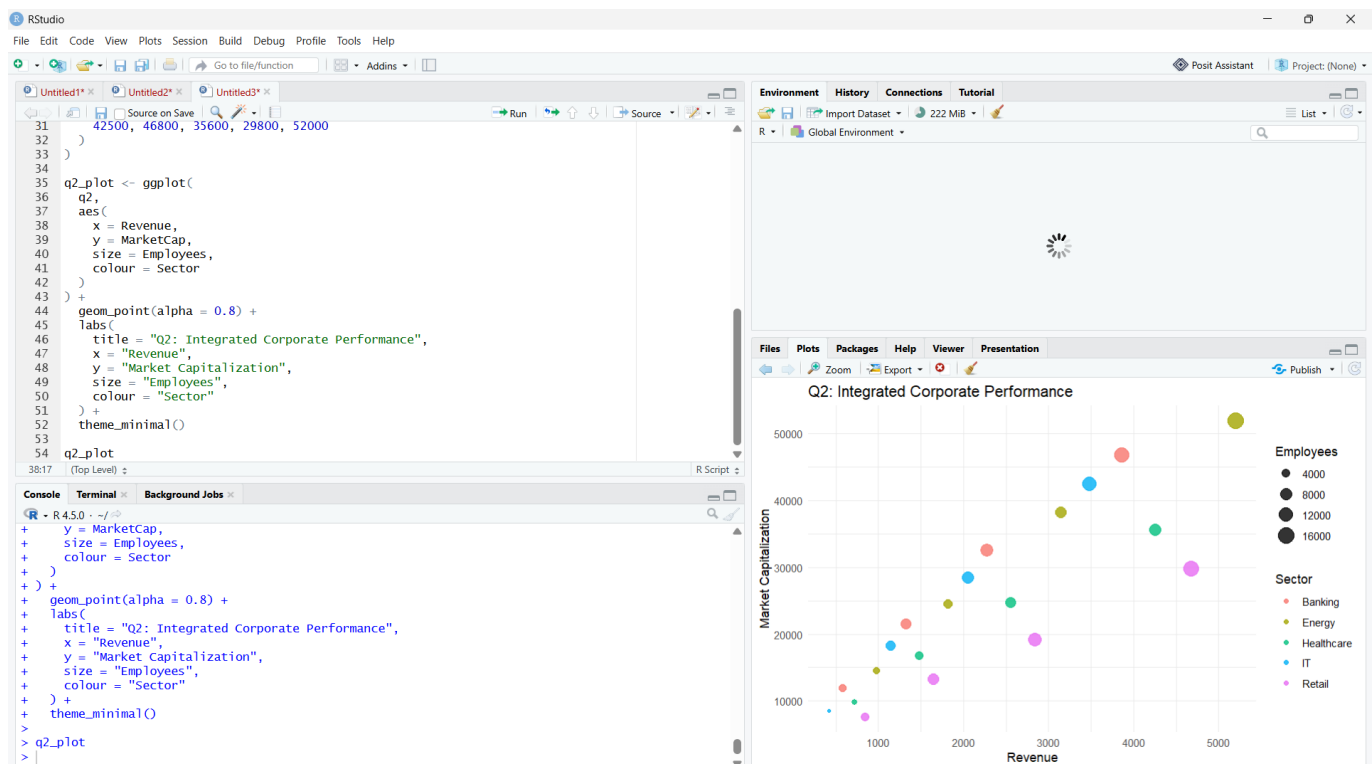
q2_plot

```

Design justification

- Position is used for the primary quantitative relationships because it is the easiest visual channel for accurate comparison.
- Colour represents Sector consistently across panels so the categorical meaning remains stable.

- Market Capitalization is mapped to point size in Panel A because it is an additional magnitude variable.
- Profit Margin is mapped to point size in Panel C, allowing profitability to be considered alongside employees and market valuation.
- The revenue bar chart is included because a dashboard should provide at least one direct comparison view rather than relying only on relationships.
- The sector boxplot provides a distribution-level view and avoids reducing an entire sector to a single average.
- All monetary variables should be labelled with their units in the final submission. The source table appears to use million-scale values for Revenue and MarketCap.



4.6 Interpretation

The integrated design is intended to support several questions at once: Which companies have larger revenue? Which companies combine higher revenue with higher profit margins? How does market capitalization vary with organizational size? Do sectors show different profitability distributions? The viewer can move from the company-level scatter

plots to the direct revenue comparison and then to the sector-level distribution without changing the meaning of the colour encoding.

conclusion

A coordinated multi-panel dashboard is more informative than a single overloaded chart for this dataset. The design preserves readability while still representing all requested variables and provides both relationship-based and distribution-based evidence.

Q3 – Cartesian vs Polar Coordinates

Problem focus: Monthly energy consumption is recorded for Coal, Gas, Solar, Wind and Hydro. The question requires visualizations using Cartesian and Polar coordinate systems and a comparison of which coordinate system provides better analytical insight.

Dataset

Variable	Type	Description
Month	Categorical / ordered	January to December
Coal	Quantitative	Monthly coal value
Gas	Quantitative	Monthly gas value
Solar	Quantitative	Monthly solar value
Wind	Quantitative	Monthly wind value
Hydro	Quantitative	Monthly hydro value

Design strategy

Two complementary views are created. The Cartesian chart uses month on the x-axis and energy value on the y-axis, making exact month-to-month comparisons straightforward. The polar chart wraps the twelve months around a circle, emphasizing the cyclic nature of the year and making seasonal patterns visually prominent.

R code – data entry and reshaping

```
q3 <- data.frame(  
  Month = factor(  
    c(  
      "Jan", "Feb", "Mar", "Apr", "May", "Jun",  
      "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"  
    ),  
    levels = c(  
      "Jan", "Feb", "Mar", "Apr", "May", "Jun",  
      "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"  
    )  
  ),  
  Wind = c(80, 95, 120, 150, 170, 185, 195, 190, 175, 150, 110, 90),  
  Hydro = c(140, 150, 160, 170, 180, 195, 205, 210, 200, 185, 165, 150),  
  Gas = c(310, 295, 305, 320, 340, 355, 360, 370, 365, 350, 335, 320),  
  Solar = c(45, 60, 90, 130, 180, 230, 250, 245, 210, 160, 90, 50)
```

R code – Cartesian visualization

```
q3_cartesian <- ggplot(  
  q3,  
  aes(x = Month, group = 1)  
) +
```

```

geom_line(aes(y = Wind, colour = "Wind"), linewidth = 1) +
geom_line(aes(y = Hydro, colour = "Hydro"), linewidth = 1) +
geom_line(aes(y = Gas, colour = "Gas"), linewidth = 1) +
geom_line(aes(y = Solar, colour = "Solar"), linewidth = 1) +
geom_point(aes(y = Wind, colour = "Wind")) +
geom_point(aes(y = Hydro, colour = "Hydro")) +
geom_point(aes(y = Gas, colour = "Gas")) +
geom_point(aes(y = Solar, colour = "Solar")) +
labs(
  title = "Q3: Energy Sources – Cartesian Coordinates",
  x = "Month",
  y = "Energy Value",
  colour = "Energy Source"
) +
theme_minimal()

q3_cartesian

```

5.5 R code – Polar visualization

```

polar_plot <- ggplot(q3_long, aes(x = Month, y = Value, colour = Source, group =
Source))
geom_line(linewidth = 1)

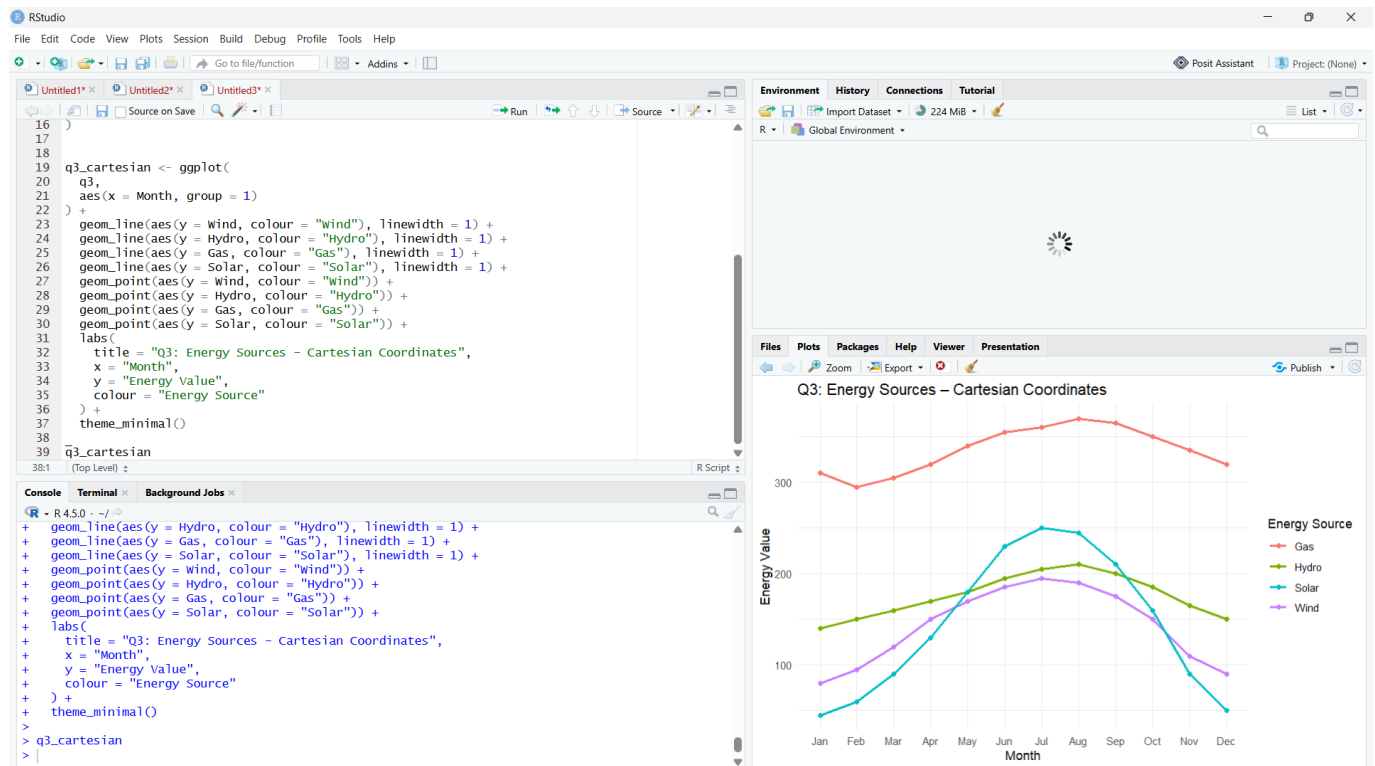
```

```

geom_point(size = 2) +
coord_polar() +
labs(
  title = "Monthly Energy Consumption – Polar Coordinates",
  x = "Month", y = "Energy Value",
  colour = "Energy Source",
  caption = "DAS0601 C01–AT3 | Q3"
) +
theme_minimal(base_size = 11)

polar_plot

```



Cartesian vs Polar comparison

Criterion	Cartesian coordinates	Polar coordinates
Month-to-month comparison	Very clear because months are arranged on a straight axis.	Less precise because angular position and radial distance are harder to compare.
Seasonality	Visible through rises and falls along the x-axis.	Very strong visual emphasis because the year forms a closed cycle.
Exact value reading	Generally easier.	Generally more difficult.
Pattern recognition	Strong for trend analysis.	Strong for cyclical/seasonal pattern recognition.
Risk of visual distortion	Low.	Radial position can make small differences harder to judge.
Best use in this dataset	Detailed analytical comparison.	Supporting view for annual seasonality.

Interpretation

- Coal remains the largest value among the five sources throughout the year in the supplied dataset.
- Solar shows a pronounced increase toward the middle of the year and then declines, making it particularly useful for demonstrating seasonality.
- Wind and Hydro change more gradually than Solar, while Gas remains at an intermediate level.
- The Cartesian chart is more suitable when the objective is accurate comparison of monthly values.

- The polar chart is useful as a complementary design because January and December are visually adjacent, reinforcing the idea of a repeating annual cycle.
- The two coordinate systems do not change the underlying data; they change how spatial relationships are represented.

Design decision

For analytical reporting, the Cartesian view should be treated as the primary chart because it supports more accurate comparisons of monthly magnitude. The polar view is valuable as a secondary visualization when the emphasis is seasonal or cyclical structure. Using both provides a stronger explanation than selecting one coordinate system without context.

Q4 – Multi-Aesthetic Visualization Design

Problem focus: A multinational retail company has annual sales information from branches across four regions. The task is to simultaneously represent Sales, Profit, Customers, Product Category and Region using suitable aesthetic mappings.

Dataset variables

Variable	Type	Visualization role
Branch	Categorical	Individual branch identifier
Region	Categorical	North, South, East or West
Category	Categorical	Electronics, Furniture, Clothing or Food
Sales	Quantitative	Revenue/sales magnitude
Profit	Quantitative	Profit or loss; includes negative values
Customers	Quantitative	Customer count

Proposed multi-aesthetic chart

Aesthetic	Mapping	Purpose
X-position	Sales	Represents branch sales magnitude.
Y-position	Profit	Shows profitability and makes losses visible below zero.
Colour	Region	Separates the four geographic regions.
Shape	Category	Adds a second categorical distinction.
Size	Customers	Shows customer scale.
Label	Branch	Identifies the branch where

		required.
--	--	-----------

Why this design fits the question

A scatter plot is particularly appropriate because Sales and Profit are both quantitative variables. Mapping them to x and y provides a direct relationship view. Region and Category are categorical variables, so colour and shape are suitable channels. Customers is quantitative, so point size can represent its magnitude. The use of a zero reference line for Profit is important because B3 and B11 have negative profit values.

R code – data entry

```
q4 <- data.frame(  
  
  Branch = paste0("B", 1:20),  
  
  Region = c(  
  
    "North", "South", "East", "West", "North",  
  
    "South", "East", "West", "North", "South",  
  
    "East", "West", "North", "South", "East",  
  
    "West", "North", "South", "East", "West"  
  
  ),  
  
  Category = c(  
  
    "Electronics", "Furniture", "Clothing", "Food",  
  
    "Electronics", "Furniture", "Clothing", "Food",  
  
    "Electronics", "Furniture", "Clothing", "Food",  
  
    "Electronics", "Furniture", "Clothing", "Food",  
  
    "Electronics", "Furniture", "Clothing", "Food"  
  
  ),  
  
  Sales = c(  
  
    450000, 620000, 180000, 350000, 920000,  
  
    550000, 725000, 1080000, 560000, 670000,  
  
    195000, 483000, 430000, 760000, 280000,
```

1150000, 510000, 690000, 220000, 1250000

),

Profit = c(

45000, 95000, -12000, 185000, 250000,

120000, 180000, 225000, 72000, 98000,

-8000, 205000, 51000, 132000, 24000,

250000, 69000, 115000, 15000, 285000

),

Customers = c(

1200, 2500, 980, 4200, 1800,

3000, 1250, 5200, 2100, 3300,

1100, 4700, 1750, 3600, 1450,

5600, 2200, 3900, 1600, 6100

)

R code – main multi-aesthetic visualization

```
q4_plot <- ggplot(
```

```
  q4,
```

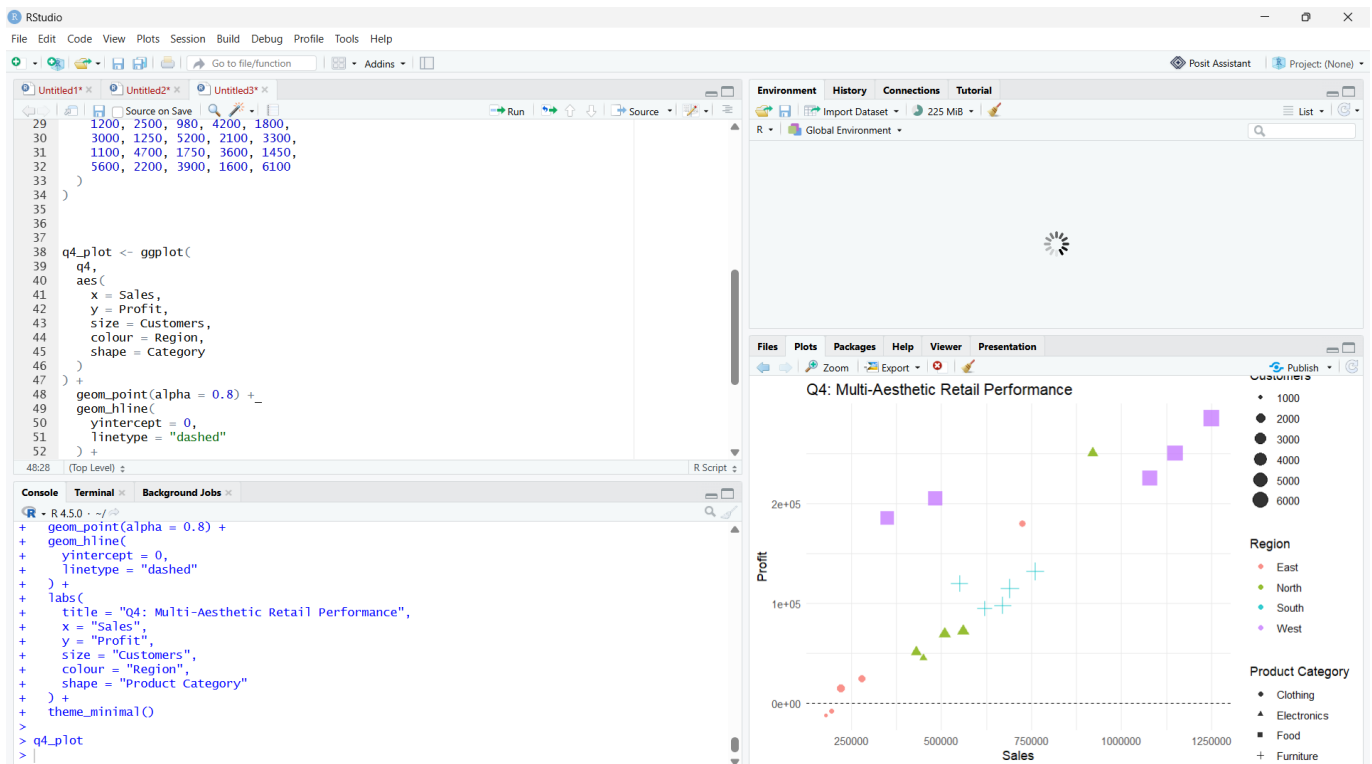
```
  aes(
```

```
    x = Sales,
```

```
    y = Profit,
```

```
    size = Customers,
```

```
    colour = Region,  
    shape = Category  
  )  
)+  
  geom_point(alpha = 0.8) +  
  geom_hline(  
    yintercept = 0,  
    linetype = "dashed"  
  ) +  
  labs(  
    title = "Q4: Multi-Aesthetic Retail Performance",  
    x = "Sales",  
    y = "Profit",  
    size = "Customers",  
    colour = "Region",  
    shape = "Product Category"  
  ) +  
  theme_minimal()  
  
q4_plot
```



Interpretation

- Sales and Profit show a broadly positive relationship: branches with larger sales generally also report larger positive profits.
- B3 and B11 have negative Profit values, so they appear below the zero line and should not be hidden or transformed away.
- Customers are encoded using point size, making high-customer branches visually prominent.
- Region is represented by colour and Category by shape, providing two independent categorical distinctions.
- The main plot is intentionally not based on a 3D chart; 2D position is easier to read and allows the remaining variables to be mapped to other aesthetics.
- The zero-profit reference line is analytically important because it separates profitable and loss-making branches.

conclusion

The multi-aesthetic scatter plot satisfies the requirement to simultaneously communicate the five requested dimensions. It balances information density with interpretability by assigning quantitative variables to position/size and categorical variables to colour/shape.

Overall Findings and Comparative Discussion

Comparison of the four design problems

Question	Main challenge	Key design solution	Main insight
Q1	Strongly skewed quantitative scale	Logarithmic x-axis + bubble aesthetics	Makes small and large projects comparable.
Q2	Many business variables	Coordinated multi-panel dashboard	Combines company, sector, profitability and valuation views.
Q3	Coordinate-system choice	Cartesian + polar comparison	Cartesian supports precision; polar emphasizes annual cyclicity.
Q4	Many quantitative and categorical variables	Multi-aesthetic scatter plot	Shows sales-profit relationship while encoding region, category and customers.

Why visualization design matters

A visualization is not simply a graph produced from data. A good design begins by identifying the analytical question, determining whether variables are quantitative or categorical, and selecting visual channels that allow the intended comparison to be made accurately. Position on a common axis is generally preferred for precise quantitative comparison, while colour, shape and size are useful secondary encodings. Transformations such as logarithmic scaling should be used when they solve a specific problem in the data rather than being applied only for appearance.

Reproducibility and quality checks

- Confirm that all row counts match the original assignment: 20 teams, 20 companies and 20 branches; 12 months for Q3.
- Check minimum and maximum values using `summary()` before plotting.
- Verify that categorical levels are spelled consistently.
- Use a zero reference line where negative profit is possible.
- Check legends for duplicated or misleading labels.
- Make sure units are shown in axis labels or captions.
- Do not crop the screenshot so tightly that the title, axes or legend disappear.
- Run the final script from the beginning in a fresh R session before submission.

Final Conclusion

The four visualizations demonstrate four different visualization-design principles. Q1 uses nonlinear scaling to handle a highly skewed variable. Q2 demonstrates integrated dashboard design for multidimensional financial data. Q3 compares two coordinate systems and explains the trade-off between analytical precision and cyclic pattern emphasis. Q4 demonstrates multi-aesthetic encoding for simultaneous quantitative and categorical analysis. Together, the solutions show how the choice of scale, coordinate system, layout and aesthetics should follow the structure of the data and the analytical purpose.

Appendix A – Full Dataset Tables

A.1 Q1 dataset

Team	LinesOfCode	Bugs	Devs	Complexity
T1	12000	35	5	2.1
T2	18000	42	7	2.4
T3	25000	60	9	2.7
T4	35000	72	10	3.0
T5	48000	88	12	3.2
T6	62000	110	14	3.5
T7	81000	125	16	3.8
T8	96000	142	18	4.1
T9	120000	165	20	4.5
T10	150000	188	22	4.8
T11	185000	205	25	5.1
T12	240000	240	28	5.4
T13	310000	280	30	5.8
T14	390000	320	34	6.2
T15	480000	365	38	6.6
T16	620000	420	42	7.1
T17	810000	490	46	7.6
T18	1050000	560	50	8.0
T19	1350000	640	55	8.5
T20	1800000	720	60	9.0

A.2 Q2 dataset

Company	Sector	Revenue	ProfitMargin	Employees	MarketCap
C1	IT	420	8.2	1500	8500
C2	Banking	580	15.6	3200	12000
C3	Healthcare	720	11.4	2100	9800
C4	Retail	850	9.3	4200	7600
C5	Energy	980	18.5	2800	14500
C6	IT	1150	20.4	5100	18200
C7	Banking	1320	17.8	6800	21500
C8	Healthcare	1480	14.2	3900	16800
C9	Retail	1650	10.5	7200	13200
C10	Energy	1820	22.1	4500	24500
C11	IT	2050	24.5	8100	28500
C12	Banking	2280	19.8	9500	32500
C13	Healthcare	2550	16.7	6200	24800
C14	Retail	2840	13.4	10300	19200
C15	Energy	3150	26.2	7200	38200
C16	IT	3480	28.1	11800	42500
C17	Banking	3860	21.5	13200	46800
C18	Healthcare	4250	18.3	8600	35600
C19	Retail	4680	15.2	14500	29800
C20	Energy	5200	30.4	16000	52000

A.3 Q3 dataset

Month	Coal	Gas	Solar	Wind	Hydro
Jan	620	310	45	80	140
Feb	580	295	60	95	150
Mar	590	305	90	120	160
Apr	605	320	130	150	170
May	640	340	180	170	180
Jun	670	355	230	185	195
Jul	690	360	250	195	205
Aug	720	370	245	190	210
Sep	700	365	210	175	200
Oct	680	350	160	150	185
Nov	650	335	90	110	165
Dec	630	320	50	90	150

A.4 Q4 dataset

Branch	Region	Category	Sales	Profit	Customers
B1	North	Electronics	450000	45000	1200
B2	South	Furniture	620000	95000	2500
B3	East	Clothing	180000	-12000	980
B4	West	Food	920000	185000	4200
B5	North	Electronics	350000	25000	1800
B6	South	Furniture	720000	120000	3000
B7	East	Clothing	250000	18000	1250
B8	West	Food	1080000	225000	5200
B9	North	Electronics	560000	72000	2100
B10	South	Furniture	670000	98000	3300
B11	East	Clothing	195000	-8000	1100
B12	West	Food	980000	205000	4700
B13	North	Electronics	430000	51000	1750
B14	South	Furniture	760000	132000	3600
B15	East	Clothing	280000	24000	1450
B16	West	Food	1150000	250000	5600
B17	North	Electronics	510000	69000	2200
B18	South	Furniture	690000	115000	3900
B19	East	Clothing	220000	15000	1600
B20	West	Food	1250000	285000	6100